

- 1 A student investigated the effect of wind speed on the rate of transpiration. The student used the flowering plant Japanese spiraea, *Spiraea japonica*.

The student used the apparatus shown in Fig. 1.1 to measure the rate of transpiration.

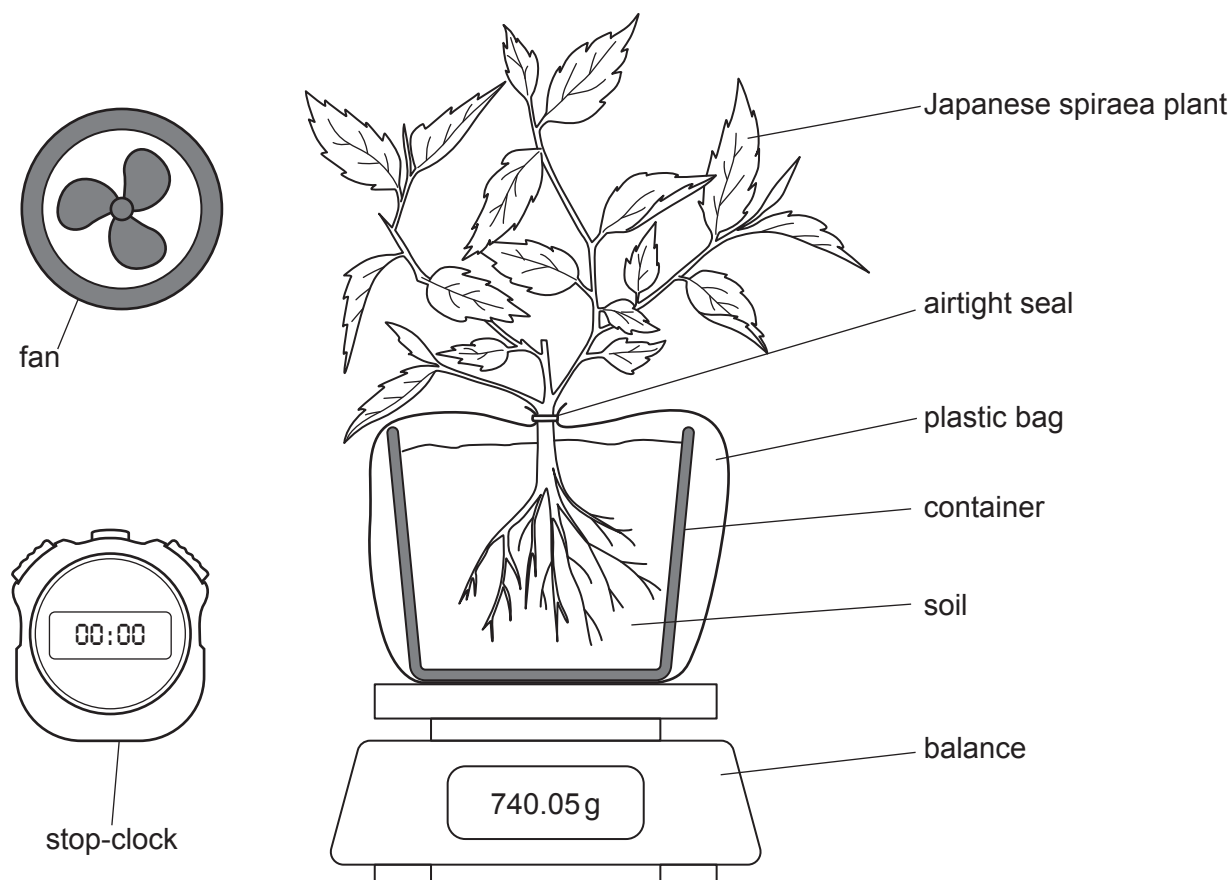


Fig. 1.1

To set up the apparatus the student:

- obtained a Japanese spiraea plant growing in a container of soil
- added 200 cm³ of water to the soil
- placed a plastic bag around the container of soil to prevent water loss from the soil
- placed the plant and container on the balance
- switched on the fan to a low setting.

The roots of the Japanese spiraea plant absorbed water from the soil. Water was carried in the xylem and water vapour was lost by transpiration from the leaves of the plant. This caused the reading on the balance to decrease during the investigation.

- (a) (i) Identify the **independent** variable in this investigation.

.....
..... [1]

- (ii)** The student carried out the investigation in a laboratory with standard laboratory apparatus.

Describe a method, using the apparatus shown in Fig. 1.1, that the student could use to investigate the effect of wind speed on the rate of transpiration by Japanese spiraea.

Your method should be set out in a logical order and be detailed enough to allow another person to follow it.

Details of how to set up the apparatus shown in Fig. 1.1 should **not** be included.

[6]

- (iii) Predict the effect of wind speed on the results of the investigation using the method you have given in (a)(ii).

..... [1]

(b) The student used a different method to measure the rate of transpiration of Japanese spiraea.

Fig. 1.2 shows the apparatus used. The leaf remained attached to the plant during the investigation.

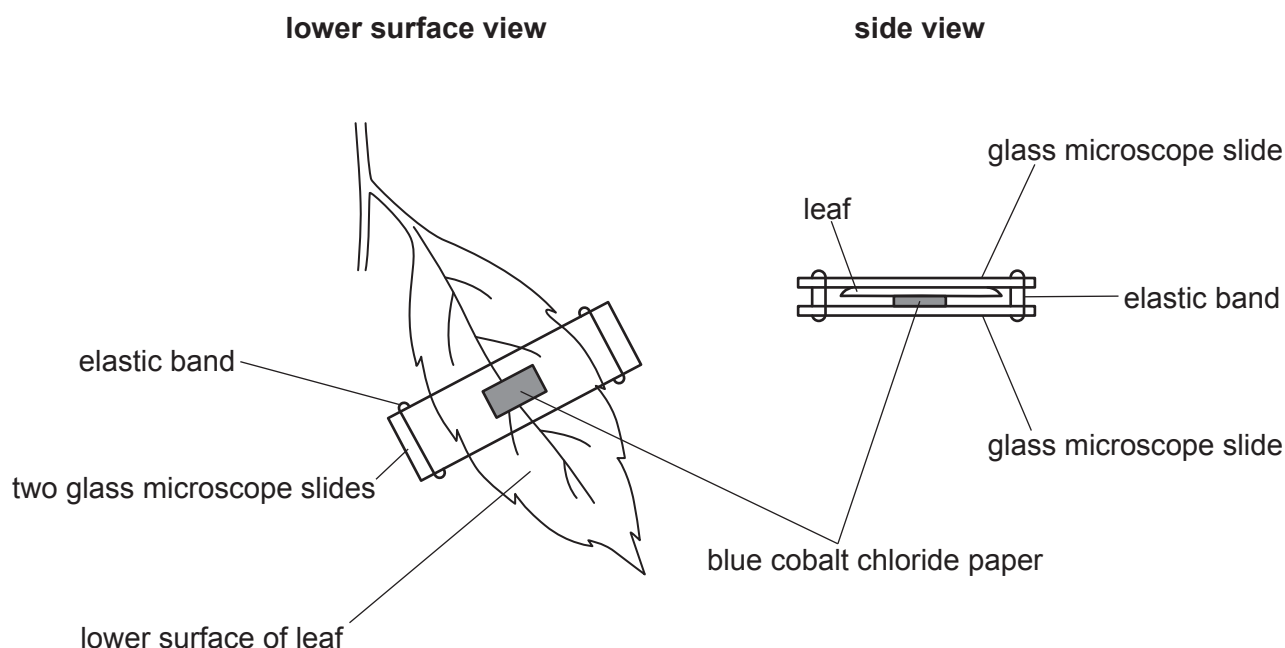


Fig. 1.2

A piece of blue cobalt chloride paper was attached to the lower surface of a leaf, as shown in Fig. 1.2. Blue cobalt chloride paper changes colour to pink if water is added.

The student measured the time taken for the blue cobalt chloride paper to change colour.

This procedure was repeated with two more leaves of the Japanese spiraea plant. The plant was kept in controlled conditions at all times.

Table 1.1 shows the results obtained using blue cobalt chloride paper.

Table 1.1

leaf tested	time taken for blue cobalt chloride paper to change colour/s
1	122
2	137
3	74

Fig. 1.3 shows the formula the student used to calculate the rate of transpiration in units of h^{-1} .

$$\text{rate of transpiration} = \frac{3600}{\text{time taken in seconds}}$$

Fig. 1.3

- (i) Using the information given in Table 1.1 and Fig. 1.3, calculate the rate of transpiration for leaf **2** of the Japanese spiraea plant.

Give your answer to **three** significant figures.

rate of transpiration = h^{-1}
[1]

- (ii) To improve the validity of the results, the student decided to measure the time taken for blue cobalt chloride paper to change colour on a greater number of leaves of the Japanese spiraea plant.

State **one** other change the student could make to the method to improve the validity of the results.

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..... [1]

[Total: 10]